1. Fundamentals of Probability Theory

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FS26 MTH345: Deep Generative Models: A Statistical Perspective

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Lab Homepage

Lecture Topics

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Lecture Topics

- This lecture briefly reviews preliminary concepts in probability.
- Useful materials include:
  - Mathematics for Machine Learning [MML] (Deisenroth et al., 2020)
  - A First Course in Probability [FCP] (Ross, 2020)
  - Mathematical Statistics and Data Analysis [MSDA] (Rice, 2007)
- Selected examples and text excerpts are reproduced from the materials for educational use only under fair use.

Outline

1. Probability

2. Conditional Probability and Event Independence

Probability

- What is probability? In our daily lives, we frequently use the term *probability* in several different contexts, including:
  1. The limit of relative frequencies
  2. The degree of belief
- For the first interpretation, consider the random experiment of tossing a coin. Suppose we perform $n = 5$ repeated trials and observe: H, T, H, T, H.
- We can construct a probability table based on relative frequencies:

Outcome	H	T
Relative Frequency	3/5	2/5

- Assuming that the coin is fair, as $n \to \infty$, the sequence of probability tables is expected to converge to:

Outcome	H	T
Limit of Relative Frequency	1/2	1/2

Probability

- **Q**: Suppose that, for any event A, the corresponding relative frequency converges to $P(A)$ as n increases. Under such an assumption, how can we show that $P(\{HH, HT, TH\}) = 3/4$?[1]
- Let $n(A)$ denote the number of trials in which event A occurs. In this discussion, the relative frequency $n(A)/n$ is assumed to converge to $P(A)$.
- Let's further assume that the coin is fair, i.e., the relative frequency of $\{HH\}$ in the long-run is the same as that of $\{HT\}$, $\{TH\}$, and $\{TT\}$ in the limit. Therefore,

$$P(\{HH\}) = P(\{HT\}) = P(\{TH\}) = P(\{TT\}). \tag{1}$$

[1] Note that we are simply using $P(\cdot)$ to denote the limit of relative frequencies and have not assumed any further axioms yet.

Probability

Exercise 1

Using the relative frequency-based definition of probability, show that $P(\{HH, HT, TH\}) = 3/4$ for a fair coin.

Probability



- For the second interpretation, someone might say:

  "This is a dog with 30% probability."

- The probability here represents our degree of confidence (0: no confidence, 1: absolute certainty), rather than the limit of relative frequencies.
- Reflecting this individual's belief, we can construct another probability table:

Outcome	Dog	Not a dog
Degree of Belief	.3	.7

This image was generated by Google Gemini.

Probability



This image was generated by Google Gemini.

Probability

- Modern probability theory is built upon fundamental assumptions (*axioms*) that unify the notion of probability across these different contexts.
- We first define the *sample space* S, the set of all possible outcomes. For example,

$$S = \{H, T\}. \tag{2}$$

- An *event* A is defined as a subset of the sample space. We then introduce the *event space* $\mathcal{A}$, which is the collection of all possible events. For example,

$$\mathcal{A} = \{\emptyset, \{H\}, \{T\}, \{H, T\}\}. \tag{3}$$

  Here, $\{H\}$ is the event where only heads is observed, and $\{H, T\}$ is the event where either heads or tails is observed (i.e., any possible outcome occurs).
- Now, the probability $P(\cdot)$ can be viewed as a mapping from $\mathcal{A}$ to $[0, 1]$. For example,

$$P(\emptyset) = 0, \quad P(\{H\}) = 1/2, \quad P(\{T\}) = 1/2, \quad P(\{H, T\}) = 1. \tag{4}$$

Probability

- Note that both the relative frequency-based and the degree of belief-based interpretations share some fundamental characteristics.
- **Probability Space:** The $(S, \mathcal{A}, P)$ is called a *probability space* if it satisfies:
  1. $P(S) = 1$.
  2. If $A \in \mathcal{A}$, then $P(A) \geq 0$.
  3. If $A_1, A_2, \cdots \in \mathcal{A}$ are pairwise disjoint (i.e., $A_i \cap A_j = \emptyset$ for any $i \neq j$), then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i). \tag{5}$$

- In the coin-tossing example,

$$P(\emptyset) = 0, \quad P(\{H\}) = 1/2, \quad P(\{T\}) = 1/2, \quad P(\{H, T\}) = 1. \tag{6}$$

- In the generated image example,

$$P(\emptyset) = 0, \quad P(\{\text{Dog}\}) = 0.3, \quad P(\{\text{Not a dog}\}) = 0.7, \quad P(\{\text{Dog, Not a dog}\}) = 1. \tag{7}$$

Probability

Exercise 2

Let $(S, \mathcal{A}, P)$ be a probability space. Show that:

1. $P(\emptyset) = 0$.
2. $P(A^c) = 1 - P(A)$ for any event $A \in \mathcal{A}$.
3. $P(A_1) \leq P(A_2)$ for any events $A_1, A_2 \in \mathcal{A}$ such that $A_1 \subseteq A_2$.

Probability



Probability

Exercise 3

Let $(S, \mathcal{A}, P)$ be a probability space. Show that, for any two events A_1 and A_2,

$$P(A_1 \cup A_2) = P(A_1) + P(A_2) - P(A_1 \cap A_2). \tag{8}$$

Probability

- When S is finite or countably infinite[2], we can first construct a probability table:

Outcome	s_1	s_2	...
$P(\{s_i\})$	p_1	p_2	...

where $p_i \geq 0$ and $\sum_{i \geq 1} p_i = 1$. Then, the function P defined by

$$P(A) = \sum_{s \in A} P(\{s\}), \tag{9}$$

is a valid probability measure (i.e., it satisfies the three axioms).

- Conversely, any probability measure defined on a finite or countably infinite sample space uniquely determines its probability table.

[2] There exists a one-to-one mapping between S and the set of natural numbers $\mathbb{N}$.

Probability

- When S is uncountably infinite, e.g., $S = [0, 1]$, we cannot construct P in a similar way.
- Note that because the sum of uncountably many positive numbers diverges, the first axiom $P(S) = 1$ cannot hold. Furthermore, some subsets of S pose many analytic challenges.
- Instead, we limit our focus to real intervals, (a, b) where $-\infty \leq a \leq b \leq \infty$.
- We then define $\mathcal{A}$ as the collection of subsets generated by countable set operations (union, intersection, and complement) on these intervals. For instance, when $S = [0, 1]$, we have:

$$(0, 1/4] \cup [3/4, 1) = (0, 1) \cap (1/4, 3/4)^c \in \mathcal{A}. \tag{10}$$

Probability

- We first define the probability of an interval as:

$$P((a,b)) = \int_a^b f(s)\,ds \tag{11}$$

for some non-negative function f satisfying $\int_{-\infty}^{\infty} f(s)\,ds = 1$. Then, the function P extended to countable unions of pairwise disjoint intervals (a_i, b_i) by

$$P\left(\bigcup_{i=1}^{\infty}(a_i,b_i)\right) = \sum_{i=1}^{\infty} P((a_i,b_i)), \tag{12}$$

forms a valid probability measure.

- For example, when $S = [0,1]$, if P is defined as

$$P((a,b)) = b - a\left(= \int_a^b 1\,ds\right), \quad \text{where } 0 \le a \le b \le 1, \tag{13}$$

and is extended to satisfy Equation (12), then P forms a valid probability measure.

Outline

1. Probability

2. Conditional Probability and Event Independence

Conditional Probability

- Consider the random experiment of tossing a coin twice. The corresponding probability table can be expressed as:

Event	{HH}	{HT}	{TH}	{TT}
Probability	1/4	1/4	1/4	1/4

- **Q:** Suppose we are informed that the first coin toss resulted in H. Do we still assign the same probability to each outcome?
- *Conditional probability* provides a principled framework for updating our probabilities based on such newly acquired information.

Conditional Probability

- Without any additional information, the probability table is:

Event	{HH}	{HT}	{TH}	{TT}
Probability	1/4	1/4	1/4	1/4

- Given that the first coin toss resulted in H (i.e., $\{HH, HT\}$ occurs), the updated table becomes:

Event	{HH}	{HT}	{TH}	{TT}
Probability				

Conditional Probability

- Without any additional information, the probability table is:

Event	{HH}	{HT}	{TH}	{TT}
Probability	1/4	1/4	1/4	1/4

- Given that the first coin toss resulted in H (i.e., $\{HH, HT\}$ occurs), the updated table becomes:

Event	{HH}	{HT}	{TH}	{TT}
Probability			0	0

Conditional Probability

- Without any additional information, the probability table is:

Event	{HH}	{HT}	{TH}	{TT}
Probability	1/4	1/4	1/4	1/4

- Given that the first coin toss resulted in H (i.e., $\{HH, HT\}$ occurs), the updated table becomes:

Event	{HH}	{HT}	{TH}	{TT}
Probability	1/2	1/2	0	0

Conditional Probability

- Without any additional information, the probability table is:

Event	{HH}	{HT}	{TH}	{TT}
Probability	1/4	1/4	1/4	1/4

- Given that the first coin toss resulted in H (i.e., $\{HH, HT\}$ occurs), the updated table becomes:

Event	{HH}	{HT}	{TH}	{TT}
Probability	1/2	1/2	0	0

- Note that the probability of $\{HH\}$ increases from $1/4$ to $1/2$. This illustrates that the event $\{HH\}$ is dependent on the occurrence of the event $\{HH, HT\}$.

Conditional Probability

- Let B be an event with $P(B) > 0$ and $Q_B : \mathcal{A} \to [0, 1]$ be the probability measure reflecting the occurrence of B. For any $A \in \mathcal{A}$:

$$\begin{aligned} Q_B(A) &= Q_B(A \cap B) + Q_B(A \cap B^c) \\ &= Q_B(A \cap B) \\ &= k(B)P(A \cap B) \quad \text{for some } k(\cdot). \end{aligned} \tag{14}$$

  By substituting $A = S$, we obtain $k(B) = 1/P(B)$. Therefore,

$$Q_B(A) = \frac{P(A \cap B)}{P(B)}. \tag{15}$$

Conditional Probability

- Let B an event with $P(B) > 0$ and $Q_B : \mathcal{A} \to [0, 1]$ be the probability measure reflecting the occurrence of B. For any $A \in \mathcal{A}$:

$$\begin{aligned} Q_B(A) &= Q_B(A \cap B) + Q_B(A \cap B^c) \\ &= Q_B(A \cap B) \\ &= k(B)P(A \cap B) \quad \text{for some } k(\cdot). \end{aligned} \tag{16}$$

  By substituting $A = S$, we obtain $k(B) = 1/P(B)$. Therefore,

$$Q_B(A) = \frac{P(A \cap B)}{P(B)}. \tag{17}$$

- **Conditional Probability**: Let A and B be two events with $P(B) > 0$. The *conditional probability* of A given B, denoted by $P(A \mid B)$, is defined as:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}. \tag{18}$$

Conditional Probability

Exercise 4

Let $(S, \mathcal{A}, P)$ be a probability space. For any event B such that $P(B) > 0$, let $Q_B : \mathcal{A} \to [0, 1]$ be defined as:

$$Q_B(A) = P(A \mid B). \tag{19}$$

Show that $(S, \mathcal{A}, Q_B)$ is also a probability space.

Conditional Probability

- How do we reflect the occurrence of multiple events?
- Suppose we are informed that the first coin toss resulted in H (i.e., the event $B_1 = \{HH, HT\}$ occurs), which updates the probability table to:

Event	{HH}	{HT}	{TH}	{TT}
Probability	1/2	1/2	0	0

- Subsequently, we are further informed that the second coin toss resulted in T (i.e., the event $B_2 = \{HT, TT\}$ occurs). Then, the updated table becomes:

Event	{HH}	{HT}	{TH}	{TT}
Probability	0	1	0	0

- How do we formulate this? There are two intuitive approaches to consider:

$$Q_{B_1}(A \mid B_2) \text{ (Sequential update)} \quad \text{and} \quad P(A \mid B_1 \cap B_2) \text{ (Joint update)}. \tag{20}$$

Conditional Probability

Exercise 5

Let $(S, \mathcal{A}, P)$ be a probability space. For any event B such that $P(B) > 0$, let $Q_B : \mathcal{A} \to [0, 1]$ be defined as:

$$Q_B(A) = P(A \mid B). \tag{21}$$

Show that, for any two events B_1 and B_2,

$$Q_{B_1}(A \mid B_2) = P(A \mid B_1 \cap B_2). \tag{22}$$

Conditional Probability

Bayes' Rule

- As in the coin example, we are often interested in updating probabilities to reflect new information.
- Suppose we have a *partition* of the sample space, $B_1, \cdots, B_n$ with $P(B_i) > 0$ such that:

$$\cup_{i=1}^{n} B_i = S \quad \text{and} \quad B_i \cap B_j = \emptyset \quad \text{when } i \neq j, \tag{23}$$

e.g., $B_1 = \{HH\}$, $B_2 = \{HT\}$, $B_3 = \{TH\}$, and $B_4 = \{TT\}$.

Bayes' Rule

- Given the occurrence of A, how do we update the probability table from

$$P(B_1), P(B_2), \ldots, P(B_n) \quad \text{where} \sum_{i=1}^{n} P(B_i) = 1 \tag{24}$$

to

$$P(B_1 \mid A), P(B_2 \mid A), \ldots, P(B_n \mid A) \quad \text{where} \sum_{i=1}^{n} P(B_i \mid A) = 1? \tag{25}$$

- **Bayes' Rule**: Let A and $B_1, \ldots, B_n$ be events where the B_i are pairwise disjoint, $\bigcup_{i=1}^{n} B_i = S$, and $P(B_i) > 0$ for all i. Then,

$$P(B_j \mid A) = \frac{P(A \mid B_j)P(B_j)}{\sum_{i=1}^{n} P(A \mid B_i)P(B_i)}. \tag{26}$$

Bayes' Rule

Exercise 6

Let A and $B_1, \ldots, B_n$ be events where the B_i are pairwise disjoint, $\bigcup_{i=1}^{n} B_i = S$, and $P(B_i) > 0$ for all i. Show that:

$$P(B_j \mid A) = \frac{P(A \mid B_j)P(B_j)}{\sum_{i=1}^{n} P(A \mid B_i)P(B_i)}. \tag{27}$$

Bayes' Rule

- Suppose three plants, B_1, B_2, and B_3, produce 50%, 30%, and 20% of a company's total output, respectively. From historical quality control data, the defect rates for the items produced by these plants are 2%, 3%, and 5%, respectively.
- **Q:** An inspector randomly selects an item from the combined output and finds it to be defective. What is the probability that the defective item was produced by each of the plants?

Bayes' Rule

- Let D denote the event that a randomly selected item is defective.

$$\begin{aligned} P(B_1 \cap D) &= P(D \mid B_1)P(B_1) = 0.02 \times 0.50 = 0.010 \\ P(B_2 \cap D) &= P(D \mid B_2)P(B_2) = 0.03 \times 0.30 = 0.009 \\ P(B_3 \cap D) &= P(D \mid B_3)P(B_3) = 0.05 \times 0.20 = 0.010. \end{aligned} \tag{28}$$

- The total probability of a defect is $P(D) = 0.010 + 0.009 + 0.010 = 0.029$.
- Applying Bayes' Rule, the posterior probabilities are:

$$\begin{aligned} P(B_1 \mid D) &= \frac{0.010}{0.029} \approx 0.345 \\ P(B_2 \mid D) &= \frac{0.009}{0.029} \approx 0.310 \\ P(B_3 \mid D) &= \frac{0.010}{0.029} \approx 0.345. \end{aligned} \tag{29}$$

Bayes' Rule

Exercise 7

Let today's weather be one of three outcomes: Sunny (S_0), Cloudy (C_0), or Rainy (R_0), with probabilities:

$$P(S_0) = 0.5, \quad P(C_0) = 0.3, \quad \text{and} \quad P(R_0) = 0.2.$$

Tomorrow's rain event (R_1) depends on today's weather as follows:

$$P(R_1 \mid S_0) = 0.1, \quad P(R_1 \mid C_0) = 0.4, \quad \text{and} \quad P(R_1 \mid R_0) = 0.7.$$

(a) Find the probability that it rains tomorrow, $P(R_1)$.

Bayes' Rule

(b) Given that it rains tomorrow, find the probability that today was sunny, $P(S_0 \mid R_1)$.

Odds

- **Odds**: The odds of an event A are defined as

$$\frac{P(A)}{P(A^c)} = \frac{P(A)}{1 - P(A)}. \tag{30}$$

- For instance, if $P(A) = 2/3$, then $P(A) = 2P(A^c)$, so the odds are 2. If the odds equal α, it is common to say that the odds are "α to 1" in favor of the event.

Odds

Exercise 8

Suppose a lottery ticket has a winning probability of $P(\text{Win}) = 0.2$.

(a) Compute the odds in favor of winning.

(b) Suppose another lottery ticket is designed such that the odds in favor of winning is $\frac{1}{99}$. Find the winning probability $P(\text{Win})$ for this ticket.

Odds

- Consider a hypothesis H that is true with probability $P(H)$[3], and suppose that new evidence E is introduced.
- Then, the odds after observing E are:

$$\underbrace{\frac{P(H \mid E)}{P(H^c \mid E)}}_{\text{Posterior odds}} = \underbrace{\frac{P(H)}{P(H^c)}}_{\text{Prior odds}} \cdot \underbrace{\frac{P(E \mid H)}{P(E \mid H^c)}}_{\text{Likelihood ratio}}. \tag{31}$$

- That is, the posterior odds of H are equal to the prior odds multiplied by the ratio of the conditional probability of the evidence given H is true to that given H is false.

[3] The probability here may be interpreted as a degree of belief.

Odds

- Suppose we have two unfair coins: Coin H, which has a 75% probability of landing on H, and Coin T, which has a 25% probability of landing on H.
- We randomly pick one of the two coins with equal probability, yielding the following prior probabilities:

Event	{Coin H}	{Coin T}
Probability	1/2	1/2

- Given that the toss result is H, we can update the odds of having chosen Coin H:

$$\underbrace{\frac{P(\text{Coin } H \mid H)}{P(\text{Coin } T \mid H)}}_{\text{Posterior odds}} = \underbrace{\frac{P(\text{Coin } H)}{P(\text{Coin } T)}}_{\text{Prior odds}} \cdot \underbrace{\frac{P(H \mid \text{Coin } H)}{P(H \mid \text{Coin } T)}}_{\text{Likelihood ratio}} = \frac{1/2}{1/2} \cdot \frac{3/4}{1/4} = 3. \tag{32}$$

- Since the posterior odds are 3 to 1, the updated probability that we chose Coin H becomes 3/4.

Event Independence

- **Independence**: Two events A and B are said to be *independent* if:

$$P(A \cap B) = P(A)P(B). \tag{33}$$

- If $P(B) > 0$, this definition is mathematically equivalent to:

$$P(A \mid B) = P(A). \tag{34}$$

That is, the occurrence of B does not change the probability of A.

Event Independence

Exercise 9

Show that, when events A and B are independent, so are A and B^C.

Event Independence

- For example, suppose we toss a fair die twice, yielding the sample space:

$$S = \{(1,1), \ldots, (6,6)\}. \tag{35}$$

- Let $A = \{(3,1), \ldots, (3,6)\}$ be the event that the first toss is a 3, and let $B = \{(1,6), \ldots, (6,1)\}$ be the event that the sum of the two tosses is 7.
- Since $P(A \cap B) = P(A)P(B) = (1/6)(1/6) = 1/36$, A and B are independent. Intuitively, any first toss leaves exactly one valid second toss, keeping the conditional probability constant.
- Now, let $C = \{(2,6), \ldots, (6,2)\}$ be the event that the sum is 8. Since $P(A \cap C) = P(\{(3,5)\}) = 1/36 \neq P(A)P(C) = 5/216$, A and C are dependent. The first toss restricts the possibilities (e.g., 1 makes a sum of 8 impossible).

Event Independence

Exercise 10

([FCP] Example 3.4e) Two fair dice are thrown. Let E denote the event that the sum of the dice is 7. Let F denote the event that the first die equals 4 and G denote the event that the second die equals 3. Is E independent of $F \cap G$?

References I

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